FITNESS DATA ANALYSIS USING SQL, PYTHON AND POWER BI

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The increasing use of fitness tracking devices has generated large volumes of health-related data such as daily steps, calories burned, activity intensity, and sleep patterns. Analyzing this data can provide valuable insights into user behavior and health trends.

The objective of this project is to analyze fitness tracker data using modern data analytics tools including SQL, Python, and Power BI. SQL was used for database management and data querying, Python (Pandas) was used for data cleaning and exploratory data analysis, and Power BI was used to create interactive dashboards.

The project demonstrates a complete data analytics pipeline that transforms raw fitness data into meaningful insights. The dashboards provide visual representations of user activity patterns and sleep behavior which can help understand health trends and improve fitness monitoring.

INTRODUCTION

Fitness tracking devices such as smartwatches and wearable fitness bands are widely used to monitor health activities. These devices collect large amounts of data including steps taken, calories burned, heart rate, sleep patterns, and physical activity levels.

Analyzing this data can help identify patterns in physical activity and lifestyle habits. With the help of data analytics tools, this information can be transformed into meaningful insights that support better health decisions.

This project focuses on analyzing a fitness tracker dataset using SQL, Python, and Power BI. The goal is to explore user activity data, understand sleep patterns, and visualize the results through interactive dashboards. By using modern analytics tools, the project demonstrates how raw data can be processed and converted into actionable insights.

OBJECTIVES :-

The main objectives of this project are:

- To analyze user activity data using SQL queries
- To perform data cleaning and exploratory analysis using Python (Pandas)
- To study the relationship between steps taken and calories burned
- To analyze sleep patterns and sleep efficiency of users
- To develop interactive dashboards using Power BI
- To generate insights that help understand user fitness behavior

DATASET DESCRIPTION

The dataset used in this project is the **Fitbit Fitness Tracker dataset** which contains information about daily physical activities and sleep patterns of users.

The dataset contains multiple CSV files representing different types of data.

Main datasets used in this project include:

- dailyActivity_merged.csv
- dailySteps_merged.csv
- dailyCalories_merged.csv
- dailyIntensities_merged.csv
- sleepDay_merged.csv
- weightLogInfo_merged.csv

These datasets contain variables such as:

- User ID
- Activity Date
- Total Steps
- Calories Burned
- Active Minutes
- Sedentary Minutes
- Sleep Duration
- Time Spent in Bed

The datasets were imported into SQL and analyzed using different tools.

Id	ActivityDa	TotalSteps	TotalDista	TrackerDis	LoggedAct	VeryActive	Moderatel	LightActive	Sedentary	VeryActive	FairlyActiv
1.5E+09	########	13162	8.5	8.5	0	1.88	0.55	6.06	0	25	13
1.5E+09	########	10735	6.97	6.97	0	1.57	0.69	4.71	0	21	19
1.5E+09	########	10460	6.74	6.74	0	2.44	0.4	3.91	0	30	11
1.5E+09	########	9762	6.28	6.28	0	2.14	1.26	2.83	0	29	34
1.5E+09	########	12669	8.16	8.16	0	2.71	0.41	5.04	0	36	10
1.5E+09	########	9705	6.48	6.48	0	3.19	0.78	2.51	0	38	20
1.5E+09	########	13019	8.59	8.59	0	3.25	0.64	4.71	0	42	16
1.5E+09	########	15506	9.88	9.88	0	3.53	1.32	5.03	0	50	31

TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this project:

Tool	Purpose
SQL (MySQL)	Database creation and data querying
Python (Pandas)	Data cleaning and analysis
Matplotlib / Seaborn	Data visualization
Power BI	Interactive dashboard creation

SQL was used to store and manage the datasets and perform analytical queries.
Python was used to clean the data and perform exploratory data analysis.
Power BI was used to create dashboards to visualize insights.

SQL DATA ANALYSIS

SQL was used to create the database, import datasets, and perform analytical queries.
Multiple tables were created to store activity, steps, calories, and sleep data.

Important SQL queries performed in this project include:

- Total steps taken by each user
- Average calories burned by users
- Activity intensity analysis
- Daily steps trends
- Active minutes analysis

Example SQL query used in the project:

```
SELECT Id, ActivityDay, AVG(StepTotal) AS AvgSteps
```

```
FROM dailysteps_merged
GROUP BY Id, ActivityDay
ORDER BY ActivityDay;
```

This query calculates the total number of steps taken by each user.

SQL queries helped identify user activity patterns and summarize important metrics.

Id	ActivityDay	AvgSteps
1503960366	2016-04-12	13162
1503960366	2016-04-13	10735
1503960366	2016-04-14	10460
1503960366	2016-04-15	9762

PYTHON DATA ANALYSIS

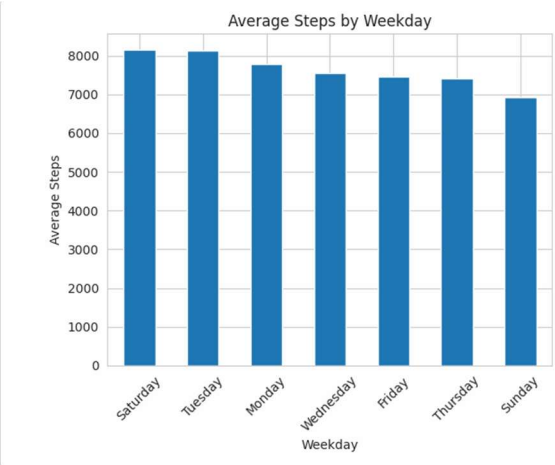
Python was used to perform exploratory data analysis using the Pandas library. The dataset was loaded into a Jupyter Notebook and analyzed using various Python functions.

The following steps were performed in Python:

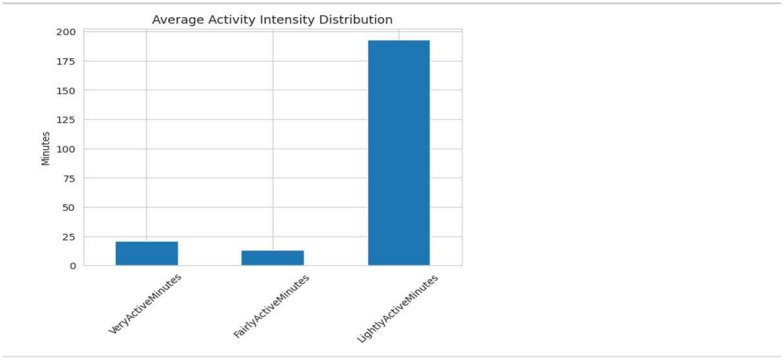
- Loading dataset using Pandas
- Data cleaning and preprocessing
- Creating new features such as weekday columns
- Analyzing activity patterns
- Calculating correlations between variables
- Creating visualizations using Matplotlib and Seaborn

Example analysis performed:

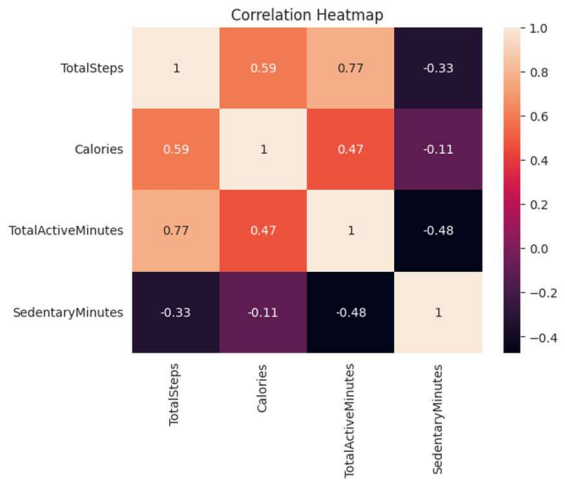
- Steps taken on different weekdays



- Activity intensity distribution



• Correlation between steps and calories burned



These analyses helped identify important relationships within the dataset.

POWER BI DASHBOARD

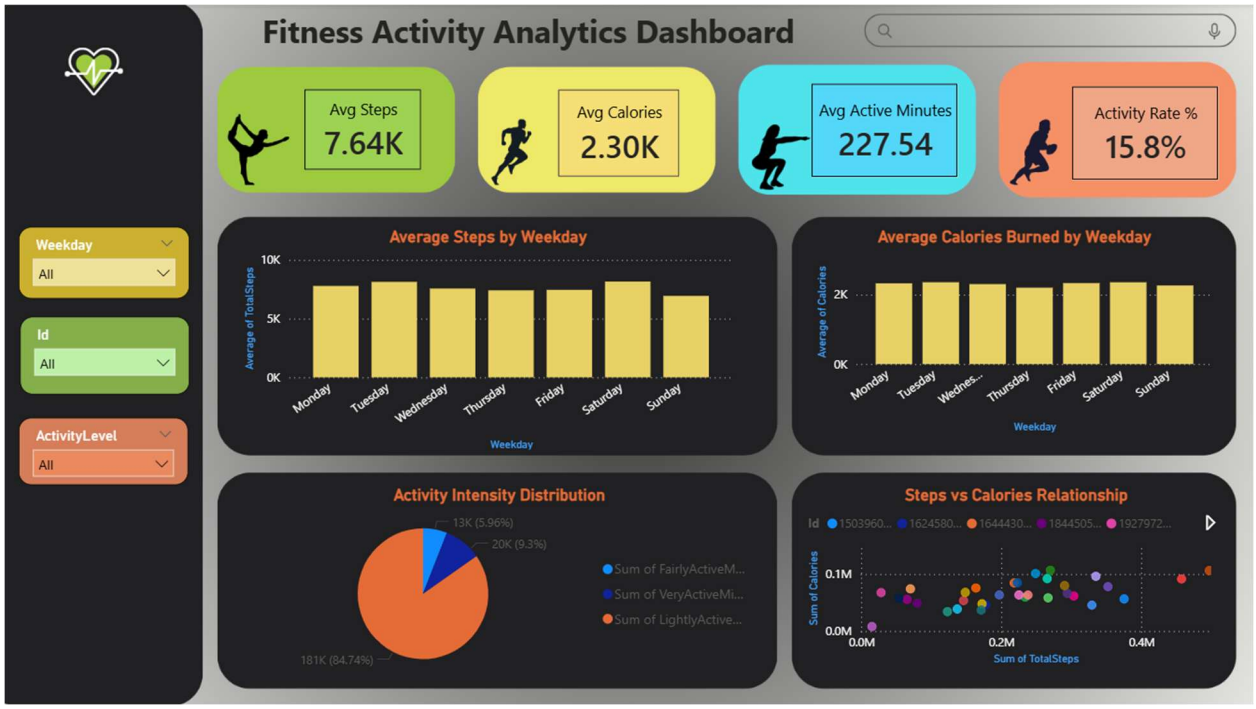
Power BI was used to create interactive dashboards to visualize the insights generated from the dataset.

Two dashboards were developed in this project.

Dashboard 1 — Activity Dashboard

This dashboard analyzes user activity trends and includes the following visualizations:

- Average Steps
- Average Calories Burned
- Average Active Minutes
- Steps by Weekday
- Calories by Weekday
- Activity Intensity Distribution
- Steps vs Calories Scatter Plot



Dashboard 2 — Sleep Dashboard

This dashboard analyzes user sleep patterns and includes:

- Average Sleep Minutes
- Average Time in Bed
- Sleep Efficiency
- Sleep Duration by Weekday
- Sleep Distribution
- Sleep vs Steps Analysis

These dashboards provide an easy way to understand activity patterns and sleep behavior.

CONCLUSION

This project demonstrates how fitness tracker data can be analyzed using modern data analytics tools such as SQL, Python, and Power BI.

SQL was used to manage and query the dataset, Python was used for exploratory data analysis, and Power BI was used to visualize the results through dashboards.

The analysis helped identify user activity patterns, calorie expenditure trends, and sleep behavior insights. The dashboards created in this project allow users to easily understand their activity and sleep habits.

Future improvements may include applying machine learning models to predict user health trends and integrating real-time fitness data for continuous monitoring.

From this project, several insights were obtained:

- average daily steps are around 7,000 to 8,000 steps
- lightly active minutes dominate user activity patterns
- steps and calories burned show a strong positive correlation
- average sleep duration is approximately 7 hours